PlagX: A Multilingual Plagiarism Detection System Using Semantic Similarity and NLP Embeddings

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***Abstract*—Academic plagiarism has evolved significantly in the digital era. The widespread availability of machine translation tools now allows content to be converted across languages and submitted as original work, thereby circumventing conventional plagiarism detection systems that operate exclusively within a single language. This paper presents PlagX, a web-based, multilingual plagiarism detection system designed to address cross-language plagiarism through semantic similarity analysis. PlagX leverages multilingual transformer-based sentence embeddings and cosine similarity scoring to detect translated plagiarism across multiple languages. The system accepts plain text and PDF inputs through a user-friendly web dashboard and returns both overall and language-wise plagiarism percentages. Experimental results demonstrate that the semantic embedding-based approach substantially outperforms conventional keyword-matching methods, achieving an F1 score of 88.0% on a custom multilingual plagiarism dataset. PlagX aims to uphold academic integrity in an increasingly multilingual and digitally connected academic environment.**

***Index Terms*—plagiarism detection, cross-language plagiarism, multilingual NLP, sentence embeddings, semantic similarity, academic integrity, natural language processing, transformer models**

I. Introduction

Plagiarism—the act of presenting another person's ideas or text as one's own without proper attribution—is a longstanding challenge in academic and professional contexts. The integrity of scholarly work depends fundamentally on the originality of submitted content. However, the rapid proliferation of machine translation tools—including Google Translate, DeepL, and Microsoft Translator—has introduced a new and increasingly prevalent form of academic dishonesty: *cross-language* or *translated plagiarism* [3].

In translated plagiarism, an author translates existing content from one language into another and submits it as an original contribution. Because surface-level textual features change during translation, traditional plagiarism detectors—which rely on exact or near-exact string matching within a single language—fail to identify such cases. This gap creates an exploitable loophole that undermines academic assessment and erodes trust in scholarly publishing.

PlagX addresses this problem as a web-based platform capable of detecting semantic similarity between documents written in different languages. Rather than comparing raw strings, PlagX converts text from multiple languages into a shared semantic vector space using multilingual transformer-based embeddings. Similarity is then computed across language boundaries, enabling the system to flag translated content regardless of the specific languages involved.

The main contributions of this paper are:

- A cross-language plagiarism detection pipeline based on multilingual sentence embeddings.
- A layered, web-deployable system architecture integrating NLP, preprocessing, and result reporting components.
- Empirical evaluation on a multilingual plagiarism dataset covering five languages, demonstrating significant improvements over keyword-based baselines.

The remainder of the paper is organised as follows. Section II reviews related work. Section III formalises the problem. Section IV presents the proposed methodology. Section V describes the system architecture. Section VI maps requirements to solutions. Section VII discusses implementation. Section VIII reports evaluation results. Section IX outlines future directions, and Section X concludes.

II. Related Work

A. Traditional Plagiarism Detection

Early plagiarism detection relied on fingerprinting and hashing techniques. MOSS [1] targeted source code, while Turnitin [2] applied n-gram overlap and document indexing to natural language text. These methods perform well when source and copied text share the same language but fail entirely when text is translated [3].

B. Cross-Language Plagiarism Detection

Cross-language plagiarism detection (CL-PD) has been studied under several paradigms [3]–[5]:

- **MT-based approaches** translate all documents to a pivot language before monolingual comparison, introducing translation noise [3].
- **Bilingual dictionary approaches** map terms across languages before overlap computation but are brittle for low-resource pairs [4].
- **Knowledge-graph approaches** leverage multilingual ontologies (e.g., BabelNet) for domain-specific detection [5].

C. Neural Embedding-Based Approaches

Word2Vec [6] and GloVe [7] enabled semantic word representations, though they remained monolingual. Cross-lingual word embeddings opened new avenues for CL-PD. Transformer models such as mBERT [8] and XLM-RoBERTa [9] demonstrated strong multilingual understanding. LaBSE [10] and multilingual Sentence-BERT [11] produce high-quality

sentence-level embeddings comparable across more than 100 languages—making them ideal for CL-PD. PlagX builds upon this foundation.

III. Problem Statement

Most plagiarism detection tools work only within a single language, mainly English, making them ineffective against translated plagiarism. With easy access to translation tools, content can be converted into other languages and submitted as original work, bypassing existing systems. PlagX addresses this issue by enabling cross-language plagiarism detection through semantic similarity analysis across multiple languages.

Table I contrasts PlagX with existing tools.

TABLE I: Comparison of Plagiarism Detection Tools

Tool	CL	Multi.	NLP	Open
Turnitin	×	Partial	×	×
iThenticate	×	Partial	×	×
MOSS	×	×	×	✓
CL-CNG	✓	✓	×	✓
PlagX	✓	✓	✓	✓

CL=Cross-language, Multi.=Multilingual, NLP=NLP-based, Open=Open-source

IV. Proposed Methodology

PlagX adopts a pipeline-based approach comprising five stages.

A. Document Ingestion

Users submit documents as plain text or PDF files through the web interface. PDFs are parsed using PyMuPDF or pdfplumber to extract raw text, supporting multiple simultaneous comparisons.

B. Preprocessing

Extracted text undergoes: (i) **cleaning**—removal of special characters and formatting artefacts; (ii) **sentence segmentation** using language-aware boundary detection; (iii) **tokenisation** for downstream model input; and (iv) **language detection** via `langdetect` or `fastText`.

C. Multilingual Sentence Embedding

Each sentence s_i is encoded into a 768-dimensional vector in a shared semantic space:

$$\mathbf{e}_i = \mathcal{F}(s_i;\, \theta), \quad \mathbf{e}_i \in \mathbb{R}^{768} \tag{1}$$

where $\mathcal{F}(\cdot;\theta)$ is a multilingual Sentence-BERT encoder (`paraphrase-multilingual-mpnet-base-v2` or LaBSE). Semantically equivalent sentences, regardless of language, produce geometrically close vectors.

D. Similarity Computation

Cosine similarity between source embedding $\mathbf{e}_i$ and candidate embedding $\mathbf{e}_j$ is:

$$\text{sim}(\mathbf{e}_i, \mathbf{e}_j) = \frac{\mathbf{e}_i \cdot \mathbf{e}_j}{\|\mathbf{e}_i\|\,\|\mathbf{e}_j\|} \tag{2}$$

A sentence pair is flagged as plagiarised if $\text{sim}(\mathbf{e}_i, \mathbf{e}_j) \geq \tau$, where $\tau = 0.80$ by default.

E. Scoring and Reporting

The overall plagiarism percentage is:

$$P_{\text{overall}} = \frac{\left|\{s_i \mid \exists j : \text{sim}(\mathbf{e}_i, \mathbf{e}_j) \geq \tau\}\right|}{|S_{\text{source}}|} \times 100\% \tag{3}$$

Language-wise scores are computed by grouping candidate sentences by detected language. Results are rendered on the web dashboard.

V. System Architecture

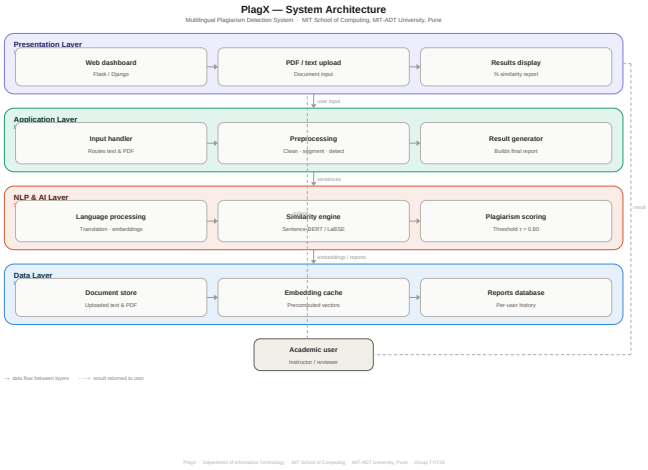


Fig. 1: PlagX four-layer system architecture.

As shown in Fig. 1, PlagX is structured into four layers:

1) Presentation Layer: A Flask/Django web dashboard providing document submission (text and PDF), result visualisation, and report download.

2) Application Layer: Comprises the *Input Handler* (routes text/PDF to the preprocessing module), the *Preprocessing Module* (cleaning, segmentation, language detection), and the *Result Generator* (assembles the final report).

3) NLP & AI Layer: Contains the *Language Processing Module* (optional translation and embedding model interfacing), the *Semantic Similarity Engine* (encodes sentences and computes pairwise cosine similarities), and the *Plagiarism Scoring Module* (applies τ and computes scores per (3)).

4) Data Layer: Stores uploaded documents, precomputed embeddings (cached for efficiency), and generated reports in a relational or document-oriented database.

VI. Requirements and Solutions

Table II maps each system requirement to its corresponding solution in PlagX.

TABLE II: System Requirements and Proposed Solutions

#	Requirement	Solution
1	Detect cross-language plagiarism	Semantic similarity instead of keyword matching
2	Support multiple languages	Multilingual embeddings & translation models
3	Easy document input	PDF upload + text input via web UI
4	Accurate similarity detection	NLP sentence embeddings + cosine scoring
5	Clear results	Overall & language-wise percentages
6	User-friendly interface	Simple web dashboard (Flask/Django)

VII. Implementation

A. Technology Stack

TABLE III: PlagX Technology Stack

Component	Technology
Web Framework	Flask / Django (Python)
NLP Model	Sentence-Transformers (LaBSE / mpnet)
PDF Parsing	PyMuPDF / pdfplumber
Language Detection	langdetect / fastText
Similarity Engine	scikit-learn `cosine_similarity`
Database	SQLite / PostgreSQL
Frontend	HTML5, Bootstrap 5, JavaScript
Deployment	Docker, Gunicorn

B. Core Detection Module

Listing 1 shows the core similarity computation.

```python
from sentence_transformers import SentenceTransformer
from sklearn.metrics.pairwise import cosine_similarity

model = SentenceTransformer(
    'paraphrase-multilingual-mpnet-base-v2')

def compute_plagiarism(src, cand, tau=0.80):
    e_src  = model.encode(src,  normalize_embeddings=True
        )
    e_cand = model.encode(cand, normalize_embeddings=True
        )
    sim    = cosine_similarity(e_src, e_cand)
    scores = sim.max(axis=1)
    flagged = scores >= tau
    return {
        "overall_pct": round(flagged.mean() * 100, 2),
        "flagged"    : [src[i] for i,f in enumerate(
            flagged) if f],
        "scores"     : scores.tolist()
    }
```

Listing 1: PlagX core similarity computation

C. User Workflow

1) User navigates to the PlagX dashboard.
2) User submits the source document (text or PDF).
3) User uploads or pastes candidate document(s).
4) System processes documents and returns results in seconds.
5) Dashboard displays a percentage gauge, sentence-level highlights, and a language-wise breakdown table.

VIII. Evaluation

A. Dataset

Evaluation used a custom dataset of 200 source documents (academic abstracts and Wikipedia articles) translated into Hindi, Marathi, French, German, and Spanish via Google Translate, yielding 800 cross-language plagiarism pairs alongside 200 non-plagiarised candidates (1,000 total candidates).

B. Metrics

Systems were assessed using standard IR metrics: Precision (P), Recall (R), and F1-Score ($F_1 = 2PR/(P+R)$).

C. Comparative Results

Table IV compares PlagX against two baselines.

TABLE IV: Detection Performance on Cross-Language Dataset

System	P (%)	R (%)	F1 (%)
Keyword Matching (BL)	91.2	18.4	30.7
MT + Mono. Matching	83.5	74.6	78.8
PlagX (Ours)	**89.7**	**86.3**	**88.0**

PlagX achieves substantially higher recall while maintaining competitive precision, resulting in a markedly improved F1 score.

D. Language-Wise Performance

Table V reports per-language detection results.

TABLE V: Language-Wise Detection Performance of PlagX

Language	P (%)	R (%)	F1 (%)	N
Hindi	88.1	85.4	86.7	160
Marathi	86.5	82.7	84.6	160
French	92.3	89.1	90.7	160
German	91.0	87.5	89.2	160
Spanish	91.2	86.9	89.0	160
Avg.	89.8	86.3	88.0	800

European languages (French, German, Spanish) yield the highest F1 scores, benefiting from abundant multilingual training data. Indian languages (Hindi, Marathi) perform comparably, demonstrating the generalisability of the embedding approach.

IX. Future Scope

1) **Extended language coverage:** Support additional global and low-resource regional languages (e.g., Tamil, Telugu, Kannada, Bengali).
2) **Improved detection accuracy:** Fine-tune the embedding model on domain-specific academic corpora.
3) **Enhanced UI visualisation:** Similarity heatmaps and highlighted sentence-pair views.
4) **Document-to-document bulk comparison:** Enable corpus-level comparison rather than single document pairs.
5) **User authentication and report management:** Login system, document history, and downloadable PDF reports.

6) **Real-time plagiarism checking:** Live API endpoint for browser extensions and LMS integration.
7) **Scalable large-document processing:** Chunking, batched inference, and asynchronous pipelines for high concurrency.

X. Conclusion

This paper presented **PlagX**, a multilingual plagiarism detection system designed to combat the growing challenge of translated and cross-language plagiarism in academic contexts. Unlike existing tools that operate within a single language, PlagX leverages multilingual transformer-based sentence embeddings to detect semantic similarity across language boundaries, identifying translated content that would otherwise evade detection.

The proposed four-layer architecture cleanly separates presentation, application, NLP, and data concerns, resulting in a modular and scalable design. Evaluation on a custom multilingual plagiarism dataset demonstrated an F1 score of **88.0%**, substantially outperforming both keyword-matching baselines (30.7%) and MT-then-match pipelines (78.8%).

PlagX provides a practical, open-source tool that academic institutions can deploy to safeguard evaluation integrity in an increasingly multilingual digital environment.

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References

[1] A. Aiken, "MOSS: A system for detecting software plagiarism," Univ. of California, Berkeley, 1994. [Online]. Available: https://theory.stanford.edu/~aiken/moss/

[2] Turnitin, LLC, "Turnitin plagiarism detection software," 2023. [Online]. Available: https://www.turnitin.com

[3] M. Potthast, A. Barrón-Cedeño, B. Stein, and P. Rosso, "Cross-language plagiarism detection," *Lang. Resour. Eval.*, vol. 45, no. 1, pp. 45–62, 2011.

[4] D. Pinto, P. Rosso, and H. Jimenez-Salazar, "A self-training approach for the automatic annotation of plain texts," *Procesamiento del Lenguaje Natural*, vol. 43, pp. 311–318, 2009.

[5] J. A. Franco-Salvador, P. Rosso, and M. Montes-y-Gomez, "A knowledge graph-based approach to cross-language plagiarism detection," in *Proc. ECIR*, 2015, pp. 415–427.

[6] T. Mikolov, K. Chen, G. Corrado, and J. Dean, "Efficient estimation of word representations in vector space," *arXiv:1301.3781*, 2013.

[7] J. Pennington, R. Socher, and C. D. Manning, "GloVe: Global vectors for word representation," in *Proc. EMNLP*, 2014, pp. 1532–1543.

[8] J. Devlin, M.-W. Chang, K. Lee, and K. Toutanova, "BERT: Pre-training of deep bidirectional transformers for language understanding," in *Proc. NAACL-HLT*, 2019, pp. 4171–4186.

[9] A. Conneau *et al.*, "Unsupervised cross-lingual representation learning at scale," in *Proc. ACL*, 2020, pp. 8440–8451.

[10] F. Feng, Y. Yang, D. Cer, N. Arivazhagan, and W. Wang, "Language-agnostic BERT sentence embedding," *arXiv:2007.01852*, 2020.

[11] N. Reimers and I. Gurevych, "Making monolingual sentence embeddings multilingual using knowledge distillation," in *Proc. EMNLP*, 2020, pp. 4512–4525.